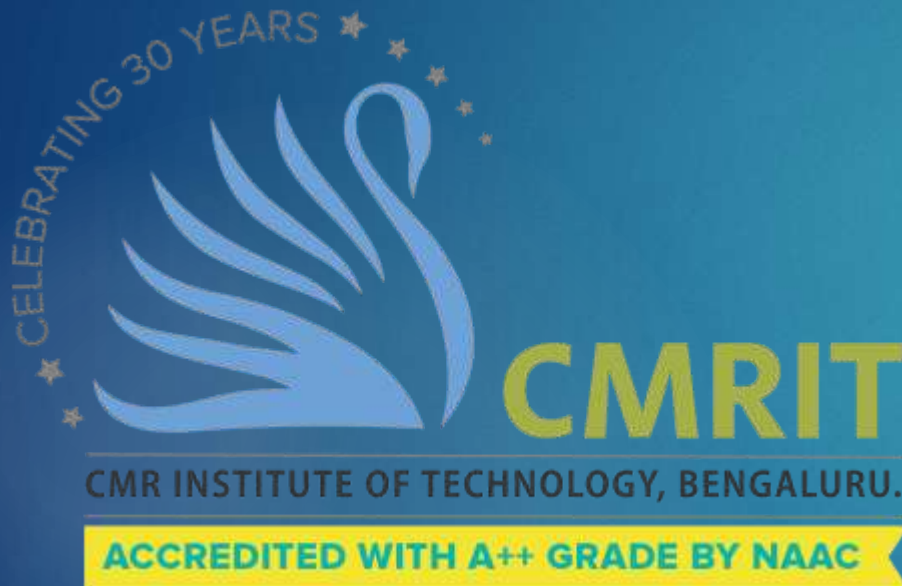


Mini Project Report - First Review

Smart Floor Cleaning Robot

Mini Project Report



BY

Namala Chandra Vedita 1CR24EC126

Prerana P Chavadi 1CR24EC157

Under the Guidance of Prof: Dr. Abinash Panda

Department of Electronics and
Communication Engineering

Date

CONTENT

- ▶ Introduction
- ▶ Objective
- ▶ Literature Review
- ▶ Methodology
- ▶ Result
- ▶ Conclusion
- ▶ Reference

INTRODUCTION

- ▶ A Smart Floor Cleaning Robot is an automated robotic system designed to clean floors with minimal human effort. The robot can move on the floor using wheels and perform cleaning using a brush and water pump mechanism.
- ▶ This project demonstrates the concept of robotics, automation, and embedded systems. The robot is built using electronic components such as NodeMCU microcontroller, motor driver module, DC motors, battery power supply, and a small water pump.
- ▶ The system helps in sweeping dust and spraying water for better cleaning. It is useful in homes, offices, hospitals, and other places where regular floor cleaning is required.

OBJECTIVE

- ▶ To design and develop a simple floor cleaning robot.
- ▶ To demonstrate the use of microcontrollers in robotics applications.
- ▶ To reduce human effort in daily floor cleaning tasks.
- ▶ To understand motor control and power supply systems.
- ▶ To integrate mechanical and electronic components into a working robotic system.

COMPONENTS USED

- ▶ NodeMCU (ESP8266) - Acts as the main controller of the robot.
- ▶ Motor Driver Module - Controls direction and speed of DC motors.
- ▶ DC Gear Motors - Used for movement of the robot wheels.
- ▶ Robot Wheels & Chassis - Provides base structure and mobility.
- ▶ Water Pump - Sprays water on the floor for cleaning.
- ▶ Brush / Fan Blade - Helps in sweeping dust particles.
- ▶ Battery, Connecting Wires, Water Pipe & Tank - Power supply and water delivery system.

WORKING MECHANISM

- ▶ The battery supplies electrical power to the NodeMCU controller and motor driver module.
- ▶ The NodeMCU acts as the brain of the robot and controls all the components.
- ▶ The motor driver module receives signals from the controller and operates the DC motors to rotate the wheels.
- ▶ A rotating brush or fan blade helps in sweeping dust particles from the floor.
- ▶ The water pump sprays water through a pipe onto the floor surface, and the combined brushing and spraying action cleans the floor as the robot moves forward.

Working Model



APPLICATIONS

- ▶ Household floor cleaning
- ▶ Offices and workplaces
- ▶ Hospitals and laboratories
- ▶ Educational robotics projects
- ▶ Smart home automation systems

CONCLUSION

- ▶ The Smart Floor Cleaning Robot is a simple and effective robotic system designed to automate the process of floor cleaning.
- ▶ The integration of NodeMCU, motors, and water pump system allows the robot to move and clean the floor simultaneously.
- ▶ This project demonstrates the practical application of robotics and embedded systems in solving real-life problems.
- ▶ With further improvements such as sensors, automatic navigation, and IoT control, the robot can become more intelligent and efficient.

FUTURE WORK

- ▶ Integration of ultrasonic/IR sensors for obstacle detection and automatic navigation.
- ▶ IoT-based remote control and monitoring via smartphone application.
- ▶ Improved battery efficiency and solar charging capability.
- ▶ AI-based path planning for fully autonomous cleaning operations.

REFERENCES

- ▶ NodeMCU ESP8266 Datasheet - Espressif Systems, 2023.
- ▶ R. Siegwart, I. Nourbakhsh, D. Scaramuzza - Introduction to Autonomous Mobile Robots, MIT Press, 2011.
- ▶ L298N Motor Driver Module - User Manual, ST Microelectronics.
- ▶ Arduino and ESP8266 Projects - <https://www.instructables.com>
- ▶ https://www.researchgate.net/publication/348127775_Automatic_Floor_Cleaning_Robot_Using_Arduino_and_Ultrasonic_Sensor
- ▶ [AUTOMATIC AND MANUAL FLOOR CLEANING ROBOT USING ARDUINO AND ULTRASONIC SENSOR - INTERNATIONAL JOURNAL OF RESEARCH AND ANALYTICAL REVIEWS](#)
- ▶ [A Comparative Study and Design for Construction of An Automatic Sanitizing Floor Cleaning Robot | PUXplore Multidisciplinary Journal of Engineering](#)